LUKE KURLANDSKI

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RESEARCH INTERESTS

My Ph.D. research develops deep learning methods for malware detection and analysis, including language models trained on executable code, adversarial attacks on malware classifiers, and detectors that generalize across platforms. Going forward, I plan to continue research at the intersection of machine learning and security, with particular interest in the robustness of large language models and their applications to defensive security.

EDUCATION

Ph.D., Computing and Information Science, Rochester Institute of Technology *Aug. 2022 – May 2027 (expected)*
- Advisors: Dr. Matthew Wright & Dr. Yin Pan
- NSF Graduate Research Fellow

B.S., Computer Science, The College of New Jersey (TCNJ) *Aug. 2018 – May 2022*
- Minors: Mathematics, Physics

PUBLICATIONS

Under Review

- **L. Kurlandski**, D. Krisiloff, S. Coull & M. Wright, "Don't Just Eat the EXE: Raw-Binary Malware Classifiers Need Structure and Scale," Under review, 2026.
- C. Xie, **L. Kurlandski**, P. Miller & N. McLaughlin, "DAUGL: Cross-Platform Malware Detection via Unsupervised Global and Local Domain Alignment," Under review, 2026.

Conference & Journal Papers

- **L. Kurlandski**, H. Berger, Y. Pan & M. Wright, "Beyond Raw Bytes: Towards Large Malware Language Models," *Network and Distributed System Security Symposium (NDSS)*, 2026.
- **L. Kurlandski** & M. Bloodgood, "Impact of Stop Sets on Stopping Active Learning for Text Classification," *IEEE International Conference on Semantic Computing (ICSC)*, 2022.
- J. Strobelt, D. Stolz, M. Leven, M. Van Soelen, **L. Kurlandski**, H. Abourahma & D. McGee, "Optical Microstructure Fabrication Using Structured Polarized Illumination," *Optics Express*, 2022.
- D. Stolz, J. Strobelt, M. Leven, **L. Kurlandski**, H. Abourahma & D. McGee, "One-Step Fabrication of Surface Relief Dot-Matrix Holograms Using Supramolecular Azopolymer Thin Films," *SPIE Practical Holography: Displays, Materials, and Applications*, 2021.
- J. Strobelt, D. Stolz, M. Leven, **L. Kurlandski** & D. McGee, "Nanoimprint Lithography for the Replication of Optical Microstructures on Azopolymer Thin Films," *IEEE Conference on Lasers and Electro-Optics (CLEO)*, 2021.

Workshop Papers

- I. Ryan, **L. Kurlandski** & N. Mathews, "Towards Moving Target Defense for IoT Malware Detection," *Workshop on Software Engineering Research and Practices for the IoT (SERP4IoT)*, 2026.
- **L. Kurlandski**, R. Mosli, Y. Pan, S. Thianphan & M. Wright, "Generating Adversarial Malware by Only Substituting API Calls," *Workshop on Rethinking Malware Analysis (WoRMA)*, 2025.

Posters & Talks

- **L. Kurlandski**, H. Berger, Y. Pan & M. Wright, "Pretrained Executable Language Models (ELMs) Excel at Malware Understanding," *USENIX Security Symposium (Poster)*, 2024.
- **L. Kurlandski**, N. Mathews, N. Hopper & M. Wright, "Modeling Stepping-Stone Flows Using Stochastic Random Walks and Black Box Neural Networks," *Western New York Image and Signal Processing Workshop (WNYISPW)*, 2024.
- **L. Kurlandski**, Y. Pan & M. Wright, "From Bytes to Insights: A New Era of Malware Detection," *Western New York Image and Signal Processing Workshop (WNYISPW)*, 2023.

- **L. Kurlandski**, Y. Pan & M. Wright, "An Adversarial Malware Evasion Attack Using Machine Translation," *AI@RIT Summit*, 2022.
- **L. Kurlandski**, "Evaluating ML Recommender Systems without Interactions Data," *ACM Technical Symposium on Computer Science Education (SIGCSE)*, 2021.

RESEARCH EXPERIENCE

Malware & Machine Learning, RIT *Aug. 2022 – Present*
Graduate Research Assistant; advised by Dr. Matthew Wright & Dr. Yin Pan.
- Language-model approaches to malware detection that move beyond raw-byte classifiers (NDSS 2026).
- Evaluating which architectural structure and scale raw-binary malware classifiers need (under review).
- Adversarial malware generation that evades detectors by substituting only API calls (WoRMA 2025).
- Collaboration on cross-platform malware detection via unsupervised domain alignment (under review).
- Collaboration on moving target defense for IoT malware detection (SERP4IoT 2026).
- Collaborations with researchers at Google, Ariel University, and Queen's University Belfast.

Cybersecurity Division, CACI International *Jun. 2021 – Aug. 2021*
Research Intern
- Researched deepfake audio detection for US Department of Defense (DoD) cybersecurity applications.

Learning Curve Group, TCNJ *Feb. 2021 – May 2022*
Undergraduate Researcher; advised by Dr. Michael Bloodgood.
- Researched active learning stopping methods for low-resource text classification (ICSC 2022).

Collective Intelligence Group, TCNJ *Jun. 2020 – Feb. 2021*
Undergraduate Research Assistant (REU, NSF-funded); advised by Dr. Monisha Pulimood.
- Built a hybrid machine learning recommendation system for academic literature (SIGCSE 2021 poster).

Photonics & Materials Science Group, TCNJ *Dec. 2018 – Feb. 2020*
Undergraduate Researcher; advised by Dr. David McGee.
- Researched optical microstructures; wrote APIs for laboratory hardware and GUI applications for experiments.

FELLOWSHIPS & AWARDS

Provost's Outstanding Graduate Student Teaching Award, RIT *2024 – 2025*
NSF Graduate Research Fellowship (GRFP) *Apr. 2024*
NDSEG Fellowship (declined in favor of NSF GRFP) *Mar. 2024*
AI Seed Fund Grant, RIT ($5,000; wrote the funded proposal) *Jan. 2024*
Goldberg-Neff Scholarship for graduate research, TCNJ *May 2022*
Phi Beta Kappa; Upsilon Pi Epsilon (CS Honor Society), TCNJ *May 2022*
Department of Computer Science First-Year Award, TCNJ *May 2019*
Four-Year Merit Scholarship, TCNJ *Aug. 2018 – May 2022*
Dean's List, all eight semesters, TCNJ *Aug. 2018 – May 2022*

TEACHING, MENTORING & LEADERSHIP

Instructor, RIT *Jan. 2024 – May 2024*
- Taught graduate research seminar on Advanced Malware Forensics.

Graduate Teaching Assistant, RIT *Aug. 2023 – Dec. 2023*
- Graduate research seminar on Human Factors in Security and Privacy.

Research Mentor, RIT *Jan. 2023 – Present*
- Mentored 19 students: 7 one-on-one independent studies (5 B.S., 2 M.S.) and 3 capstone teams (12 B.S.).

Lead Undergraduate Researcher, TCNJ *Aug. 2021 – May 2022*
- Student lead among undergraduate researchers in the Learning Curve Group.

Founder, Kappa Delta Rho Nu Gamma Chapter, TCNJ *Jan. 2019 – May 2022*
- Established the Nu Gamma Chapter of the Kappa Delta Rho fraternity at TCNJ.

GED Instructor, Oasis – A Haven for Women and Children, Paterson, NJ *Nov. 2017 – May 2018*
- Taught math, science, and English to adults preparing for the General Educational Development (GED) exam.

SKILLS

ML:	PyTorch, Hugging Face (Transformers, Datasets), scikit-learn, TensorFlow, Captum
Security:	IDA Pro, Ghidra, radare2, OllyDbg, pefile, LIEF, Capstone
Languages:	Python, C, C++, Rust, Java, R, MATLAB, Mathematica
Systems:	Linux, SLURM / HPC clusters, Git, LaTeX